Bihar Engineering University, Patna

End Semester Examination-2023

Semester-VI

Course: B. Tech ***Time: 03 Hours***

Code: 100602 ***Subject: Computer Networks*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.
(v) Symbols used (if any) have their usual meanings.

Q.1 ***Answer any seven question of the following:*** **[2 x 7 = 14]**

(a) In an optical fiber, the inner core is_____ the cladding.
(i) denser than (ii) less dense than
(iii) the same density as (iv) another name for

(b) Sliding window protocol is used for
(i) error control
(ii) session control
(iii) flow control
(iv) concurrency control

(c) Connection-establishment involves a _____-way handshake in TCP.
(i) one (ii) two
(iii) three (iv) four

(d) Which the following is an inter-domain routing protocol?
(i) RIP (ii) OSPF
(iii) BGP (iv) Both (a) & (b)

(e) Phase transition for each bit is used in
(i) NRZ encoding (ii) Manchester encoding
(iii) Carrier modulation (iv) Amplitude modulation

(f) Which one of the following is a valid host for network 192.168.4.32/68?
(i) 192.168.4.39 (ii) 192.168.4.50
(iii) 192.168.4.47 (iv) 192.168.4.31

(g) UDP is
(i) connection oriented
(ii) connection less
(iii) both (a) and (b)
(iv) none of these.

(h) The protocol that maps a physical (MAC) address to the corresponding logical address is
(i) ARP
(ii) RARP
(iii) ICMP
(iv) IMAP4

(i) Process to Process delivery is the function of_____ layer.
(i) Transport (ii) Network
(iii) Physical (iv) none of these

(j) Which channel access method is used in IEEE 802.5 networks?
(i) CSMA/CD
(ii) Token bus
(iii) Token ring
(iv) All of these

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Q.2 (a) Differentiate between circuit switching and packet switching. [7]

(b) Given a 10 bit sequence 1010011110 and a divisor 1011. Find the CRC. Check your answer. [7]

Q.3 (a) Write down the similarities and differences between OSI and TCP/IP model. [7]

(b) Explain leaky bucket and token bucket algorithm. [7]

Q.4 (a) Find the expressions for average delay and throughput for both pure ALOHA and slotted ALOHA. Compare their performances as well. [7]

(b) Why are medium access control techniques required? Discuss three popular medium access control techniques in brief. [7]

Q.5 An ISP is granted the block 80.70.56.0/21. The ISP needs to allocate addresses for two organizations each with 500 addresses, two organization each with 250 addresses, and three organizations each with 50 addresses. [14]

(a) Find the number and range of addresses in the ISP block.

(b) Find the range of addresses for each organization and the range of unallocated addresses.

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Q.6 (a) How do guided media differ from unguided media? Name the advantages of optical fiber over twisted-pair and coaxial cable. [7]

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(b) Explain how sliding window protocol works at data link layer. [7]

Q.7 (a) In Go-Back-N ARQ show the window size should be $< 2^m$. [7]

(b) We know that there are 2 types of address scheme (i.e. IP address and MAC address) in computer networks. Explain what is the need of having two types of addresses. What will happen if we use only IP address in computer networks? [7]

Q.8 (a) Discuss CSMA/CA with the help of a flowchart. [4]

(b) Why CSMA/CD is not implemented in WLAN? [3]

(c) Why acknowledgement is numbered in stop & wait protocol? Discuss the situation when unnumbered acknowledgements can create confusion in the sender receiver end. [4]

(d) Describe 802.3 header formats. Why padding is required? [3]

Q.9 Write short notes on ***any two*** of the following: [7×2=14]

(a) FTP

(b) Firewall

(b) VLAN

(c) NAT

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